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Open-Ended Technological Evolution: The Co-Evolution of Invention and Cognitive Technologies

Mathieu Charbonneau

Africa Institute for Research in Economics and Social Sciences, Université Mohammed VI Polytechnique

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Abstract

The cumulative evolution of technology has proven central to our species' ecological success, allowing for cultural rather than biological adaptation to environmental challenges. While cumulative improvement explains how specific technological traditions can get increasingly better at solving pre-existing adaptive problems, it remains fundamentally an optimization process, one which halts when an optimal solution is found. Yet, humans are also capable of open-ended or evolvable technological change, that is, we have the capacity for generating novel and useful technological solutions for an ever-expanding set of increasingly complex problems. How novel problems of increasing complexity are accessed, however, remains an open issue. Here, I argue that human open-ended technological evolution emerges from the cultural evolutionary bootstrapping of our inventive capabilities through cognitive technologies. By inventing technologies that enhance our cognitive capabilities, we become able to invent technologies that would have been impossible to design using only our core (noncultural) cognitive abilities. These inventions include further empowering cognitive technologies, creating a feedback loop through which inventors become increasingly capable of making themselves even more capable inventors. I propose a model for how the cultural evolution of increasingly sophisticated cognitive technologies enables access to previously unreachable invention problems, driving open-ended technological change. This process differs from cumulative optimization as it involves expanding the range of problems that can be solved (evolvability) rather than optimizing solutions to existing

Correspondence should be sent to Mathieu Charbonneau, Africa Institute for Research in Economics and Social Sciences, Mohammed VI Polytechnic University – Rabat Campus, Rocade Rabat-Salé, Rabat 11103, Morocco. E-mail: mathieu.charbonneau@um6p.ma

problems (adaptation). This paper contributes to our understanding of human technological uniqueness by identifying a mechanism enabling open-ended cultural evolution.

Keywords: Open-ended technological evolution; Invention; Cumulative cultural evolution; Cognitive technology

1. Introduction

An often-claimed hallmark of our species' distinctiveness is our capacity for inventing technological solutions to adaptive problems and improving upon those solutions over intergenerational time, a process called *cumulative culture* (Mesoudi & Thornton, 2018; Tomasello, 1999). While recent evidence now points in the direction that nonhuman animals may be capable of such a process (e.g., Sasaki & Biro, 2017), nonhuman traditions seem constrained to bettering solutions to specific problems that, once optimized, stop improving (Derex, 2022; Winters, 2019). In contrast, humans are capable of continuous, apparently unbounded cumulative culture, where optimal solutions to local problems are reached, but where new and increasingly complex problems are also introduced, with new, often more complex solutions found, and so on (Charbonneau, Strachan, & Winters, 2023; Winters & Charbonneau, 2025). Just what best explains this open-endedness remains an open question (Borg et al., 2024; Charbonneau, 2015; Derex, 2022).

While the idea that our capabilities for cultural evolution can themselves evolve culturally is not new—an idea known as culture-culture co-evolution (Birch & Heyes, 2021; Muthukrishna & Henrich, 2016)—, most work focuses on how mechanisms of cultural learning and their transmission fidelity can culturally improve. However, getting increasingly better at transmitting and preserving complex solutions does not explain how we manage to access ever more complex and diverse problems. Here, I propose that explaining how human culture—specifically technological culture—can evolve in an open-ended manner requires us to focus on how human *inventive capabilities* can themselves evolve culturally. However, invention—both as a process and as a driver of technological evolution—is understudied within the field of cultural evolution (Charbonneau, 2015). Indeed, its mechanics are often black-boxed under vague umbrella terms such as copying errors, serendipitous discoveries, or at best as a form of individual learning or guided variation (Muthukrishna & Henrich, 2016).

This paper contributes to our understanding of open-ended technological evolution by arguing that human open-ended (or evolvable) technological evolution is made possible by the evolutionary bootstrapping of our *inventive capabilities*. One way by which we can expand the range of inventions we are capable of is through the development of increasingly empowering cognitive technologies. More precisely, as new cognitive technologies are invented, these allow the invention of technologies that could not have been invented otherwise. Among those inventions are other empowering cognitive technologies, allowing our species to engage in a feedback loop through which we are getting smarter at making ourselves even smarter.

The argument is structured as follows. In Section 2, I clarify reasons why cumulative optimization fails to provide a mechanism for open-ended technological evolution and propose

the alternative that what is required is a cultural evolutionary process capable of improving our inventive capabilities. In Section 3, I discuss the idea that cultural evolution can change and improve its own underlying mechanisms—that of culture-culture coevolution. However, most work has focused on how transmission can be culturally improved, which I argue fails to provide a way to open-ended evolution. In Section 4, I begin to elaborate a theory of technological evolvability by providing an understanding of invention as a process of design problem-solving. This, in turn, allows me to show, in Section 5, that cognitive technologies provide novel means to solve invention problems by providing new cognitive operations to inventors. This effectively allows inventors to access and solve invention problems that were, until then, beyond their reach. Finally, in Section 6, I propose an understanding of the open-ended evolutionary process by which culture can evolve more powerful and further reaching means for invention, which engaged our species in a positive feedback loop by which we have invented and continue to invent ourselves means to become ever smarter inventors.

2. Explaining cultural evolvability

Human-specific or not, cumulative culture has been a central driver of our species' ecological success (Boyd & Richerson, 1996; Henrich, 2016). The process has allowed our ancestors to evolve means to adapt to local ecological challenges culturally rather than biologically, bringing our species to successfully invade most ecosystems on Earth. As an intergenerational process, cumulative culture would have produced adaptive solutions to ecological problems that go beyond what any individual, however smart, could have devised on its own (Tennie, Call, & Tomasello, 2009; Tomasello, 1999).

The concept of cumulative culture is polysemic and multifaceted (Mesoudi & Thornton, 2018), with several evolutionary processes falling under its label. Here, I contrast my proposal to two cultural evolutionary processes typically understood as cumulative. The first is the process of intergenerational improvement of technological solutions to existing problems, which I refer to as "cumulative optimization" for short. In the next section, I engage with the cumulative process of culture-culture coevolution of cultural transmission. The contrastive analysis I propose does not aim to reject these cumulative mechanisms, but merely to point out their limitations in providing an explanation of open-ended technological change.

As a process, cumulative optimization can be schematized in the following way (Charbonneau, 2015; Tomasello, 1999):

- CC1 An individual invents, on their own, some new kind of technology to solve an adaptive problem. (invention from scratch)
- CC2 Knowledge about how to produce and use the technology then spreads through the population and further down the generations, forming a tradition. (descent)
- CC3 Downstream, another individual modifies the now traditional technology by improving it. (modification)
- CC4 The improved version of the technology then spreads through the population and further down the generations. (descent with modification)

CC5 The process of modification and descent (steps CC2 to CC4) is repeated throughout intergenerational overturn, leading to increasingly efficient and adaptive technological traditions. (cumulative optimization)

Humans would be capable of cumulatively improving technological traditions due to their evolved cognitive capacity for high-fidelity social learning (descent, steps CC2 and CC4), which would act as a "ratchet" in preserving useful modifications (Tomasello, 1999). In contrast, our capacity for inventing and improving technologies (steps CC1 and CC3) would not require any form of sophisticated cognition, acting mostly through chance discovery and the rare copying errors creeping in our highly faithful, yet ultimately imperfect social learning (Boyd, Richerson, & Henrich, 2011; Henrich, 2016; Muthukrishna & Henrich, 2016; but see Osiurak & Reynaud, 2020; Charbonneau, 2020). All and all, this process is one of optimization: given some ecological or adaptive problem (e.g., producing an effective hunting spear), the process can incrementally lead to a (local or global) optimal solution to that problem (e.g., maximizing the aerodynamism of the spear) (Derex, 2022).

While the process of cumulative optimization can explain how specific technological traditions improve in solving some pre-existing adaptive problems, the process has its limitations (Charbonneau, 2015; Derex, 2022; Winters, 2019). Most importantly, by focusing only on how *existing* problems are solved, the optimization process does not explain how the problems it solves were accessed in the first place (Charbonneau et al., 2023). Indeed, within the optimization process, novelty only concerns solutions to pre-existing problems (steps CC1 and CC3). Cumulative optimization is a solution-finding process, not a problem-finding one. Additional mechanisms are required to explain the origins of novel problems (Charbonneau et al., 2023; Mesoudi & Thornton, 2018).

As a solution-finding process, cumulative optimization has an inbuilt stopping condition, with populations inevitably stalling within what Winters (2019) calls "optimization traps": having found a (local or global) optimal solution to available adaptive problems, there are no more improvements for the optimization process to make, leading it to stall (Charbonneau, 2015; Charbonneau et al., 2023; Winters & Charbonneau, 2025). This is supported by the repeated observations that improvements of human technological traditions solving a same specific problem (e.g., the use of water turbines for energy production) often decelerate to a stall (Smil, 2019; see also Mesoudi, 2011).

Yet, it seems that humans are also capable of open-ended or evolvable technological change, that is, of an ongoing capacity for generating novel and useful technological variations for an ever-expanding set of problems, thereby escaping optimization traps by effectively moving on to new problems (Charbonneau, 2015; Charbonneau et al., 2023; Winters, 2019; Winters & Charbonneau, 2025). Given the existence of such optimization traps, the continued evolvability of technologies is by no means inevitable. Here, I suggest that a capability for open-endedness is, at least in part, a historical, perhaps path-dependent outcome (Wimsatt, 2013, 2019; Winters & Charbonneau, 2025; but see Koppl et al., 2019 for a view that open-ended technological evolution is inevitable). The capacity for continuously expanding the range of problems we solve and the complexity of the solutions we invent, together with our capacity to adapt existing technologies to serve new purposes, goes beyond the incre-

mental improvement and stopping condition typical to cumulative optimization (Charbonneau et al., 2023; Winters & Charbonneau, 2025). Open-ended technological evolution, therefore, calls for a different explanation (Borg et al., 2024; Charbonneau, 2015; Derex, 2022; Mesoudi & Thornton, 2018; Winters, 2019; Winters & Charbonneau, 2025), one capable of identifying those mechanisms allowing human populations to escape optimization traps.

It is important to keep in mind that cumulative optimization and the process of invention bootstrapping developed here are not opposed to one another. In fact, I take them to be complementary: while the cumulative optimization process is constrained by optimization traps, the bootstrapping process developed here provides a means for populations to circumvent these constraints, in turn allowing both processes to work in pair. This being said, the two processes explain very different cultural evolutionary phenomena. Cumulative optimization is a solution-focused process that leads populations to adapt to their local environment; it explains *adaptation* through a (selection-based) hill-climbing process. In contrast, the process of invention bootstrapping developed here does not deal with adaptation and hill-climbing but rather aims to explain the increasing diversity and ever-growing complexity of problems that human populations access and settle to solve. In other words, it aims to provide an explanation of the *evolvability* of human technology (Lala & O'Brien, 2023).

In order to develop a clear account of the evolvability of human technology, we need to go beyond the mainstream view of invention as a means for improving technological traditions, that is, making them better at solving extant problems. Invention is also a process allowing us to access novel problems and, under the effect of novel cognitive technologies, invention can increase its reach toward novel, more complex problems. This co-evolutionary dynamics of invention and its products provides a new mechanics by which optimization traps can be avoided and open-ended technological evolution made possible (see also Winters & Charbonneau, 2025).

3. Culture-culture coevolution

The idea that the cultural evolutionary process can change and even improve its own mechanics through the effect of culturally evolved cognitive innovations is by no means a new one (e.g., Dennett, 2017; Donald, 1991; Goody, 1977; Heyes, 2018). Muthukrishna and Henrich (2016) refer to this feedback loop, where cultural innovations can impact the processes of cultural change that made them possible in the first place, as "culture-culture coevolution."

Research on the culture-culture feedback loop has mostly focused on how the means for the transmission of cultural knowledge (steps CC2 and CC4) have improved over time. For instance, writing systems, a culturally evolved technology, can increase the capabilities of individuals to diffuse (e.g., asynchronous communication vs. synchronous oral communication [Morin, 2022]) and retain (e.g., permanent material recording of knowledge vs. transient oral exposition) knowledge, and increase the faithfulness by which such knowledge is transmitted.

The (cultural) cumulative improvement of social learning mechanisms, while crucial to explain our species' ability to culturally transform our cognitive and cultural capabilities, does not provide a means to produce open-ended evolutionary change. For instance, Birch and Heyes (2021) propose a 5-step scenario examining how culturally evolved cognitive mechanisms (or cognitive techniques, see below) may have evolved to make cumulative optimization increasingly efficient at retaining adaptive solutions. Culturally improved transmission, they argue, allows faster cumulative change by ensuring that rare improvements are better retained and further diffused. However, faster cultural transmission at best reduces the time needed for reaching an optimal solution and stall there, that is, it does not provide the cumulative optimization process a way out of optimization traps. What is needed instead is a mechanism explaining how we culturally evolve means to access novel and increasingly complex problems, and solve them (i.e., how steps CC1 and CC3 can be culturally improved).

To the best of my knowledge, there is no systematic study of the role of cognitive technologies in expanding the abilities involved in the invention and modification of technologies, and how such changes impact cultural evolutionary dynamics. Some work focus on how rates of invention can increase, culturally—that is, just how easy (or probable) it is to invent something new (e.g., Kolodny, Creanza, & Feldman, 2016)—, but at best these provide yet another means for cumulative optimization to solve existing problems faster (by producing more variants, increased rates of invention make it more likely that a working improvement is found). Just how the cultural evolution of invention provides an increase in evolvability of technology—that is, *which* inventions or improvements are made possible at any time (Charbonneau, 2015; O'Brien & Lala, 2023)—remains generally unaddressed.

4. Invention as a design problem-solving process

In order to better understand how inventive capabilities can culturally evolve, it is important to have a clearer idea of what invention as a process consists of. Here, I approach invention as a problem-solving process, specifically one of designing new means to solve problems.

A problem is a situation where an individual, or group of individuals, (1) are set in some specific state of affairs (the starting conditions), (2) have access to a limited range of available resources and means (or operations) for using those resources, and (3) aim to reach some desired state of affairs (the goal or task) (Newell & Simon, 1972). Problem-solving is an activity that consists of finding a way, from the starting conditions, to reach the desired state (the goal), using only those means available to the problem-solver, that is, to provide a solution to the problem. For example, in order to determine which of two clumps of gold is the most valuable (task), someone lacking any tools (starting conditions) could hold each clump in different hands (means) and assess which one feels heavier (means) and, from this comparison, reason to which clump contains more gold (under the heuristic that apparent weight positively correlates with quantity, a means), and then reach a conclusion as to which clump is more valuable (solution).

Means for problem-solving consist of the technologies, materials, actions, and cognitive capabilities available to find solutions to a problem. At any time, the set of available means

can be expanded by the addition of new ones. In the case of noncognitive technologies, new or improved technologies can add to the repertoire of means available for solving problems, therefore, bolstering the problem-solving capabilities of individuals. For instance, the invention of levers, ropes, and pulleys has enabled the design of grander scale buildings, architectural problems that could not be solved without the aid of such technologies. Means for solving problems also include the cognitive processes one can deploy to organize actions over materials and existing tools (see Section 5). Inventing a new cognitive technology, then, can add to the repertoire of cognitive operations of those who use the technology, allowing new possibilities in finding solutions to their problems.

Invention is a special case of problem-solving in that it consists of *designing* (a cognitive activity) a new means that can later be implemented (e.g., materially produced) and used to solve some other problems. Shortly put, inventing consists of using available means to produce new means. For instance, if one can evaluate the weight of two clumps of gold by holding them in one's hands—finding a solution using only one's core ability to compare weights—, inventing a mechanical weighing scale consists of designing a new means for evaluating the relative mass of those valuable clumps. In the context of technological evolution, inventing is an activity where one designs a new technology (e.g., a mechanical weighing scale), or an improvement over an already existing technology (e.g., a more precise weighing scale), thereby expanding the repertoire of possible means exploitable to provide solutions to problems.

All inventions are equal in that they provide new means to solve problems, but some inventions offer more: they provide means to solve problems that could not have been solved otherwise (Charbonneau, 2015). At any point in time, the specific means available to solve a problem are limited, and these shape the range of problems one can find solutions to. Problems for which solutions can be devised using only available means are *within reach* to inventors (Fig. 1A). For instance, determining which of two clumps of gold is heavier (and, therefore, more valuable) than the other can be done using only core processes (see above). In contrast, other problems are *beyond reach* to inventors given a limited set of means (Fig. 1A). This occurs when the available means for problem-solving are inadequate to solve those problems. For instance, before the invention of the balance scale, determining the precise quantitative difference in weights of two objects was beyond reach to individuals using only their core capacities.

Increasing the number of available means through invention can help in better solving existing problems. However, it does not necessarily entail that problems beyond reach can be solved. Means that simply serve as improvements over pre-existing ones (e.g., designing a longer-lasting pen or less cumbersome reading glasses) may offer better solutions to existing problems, but such improvements alone do not imply that new problems—problems that are beyond-reach to the inventor—now become accessible. Cumulative optimization can explain how optimal solutions can be found for existing problems through a process of descent with modification, leading to optimization traps where no further improvements are possible (see above). But it does not account for how problems beyond reach are accessed and solved, that is, it lacks the capacity for explaining technological evolvability (Charbonneau et al., 2023; O'Brien & Lala, 2023). To explain why technological change is open-ended, it is necessary

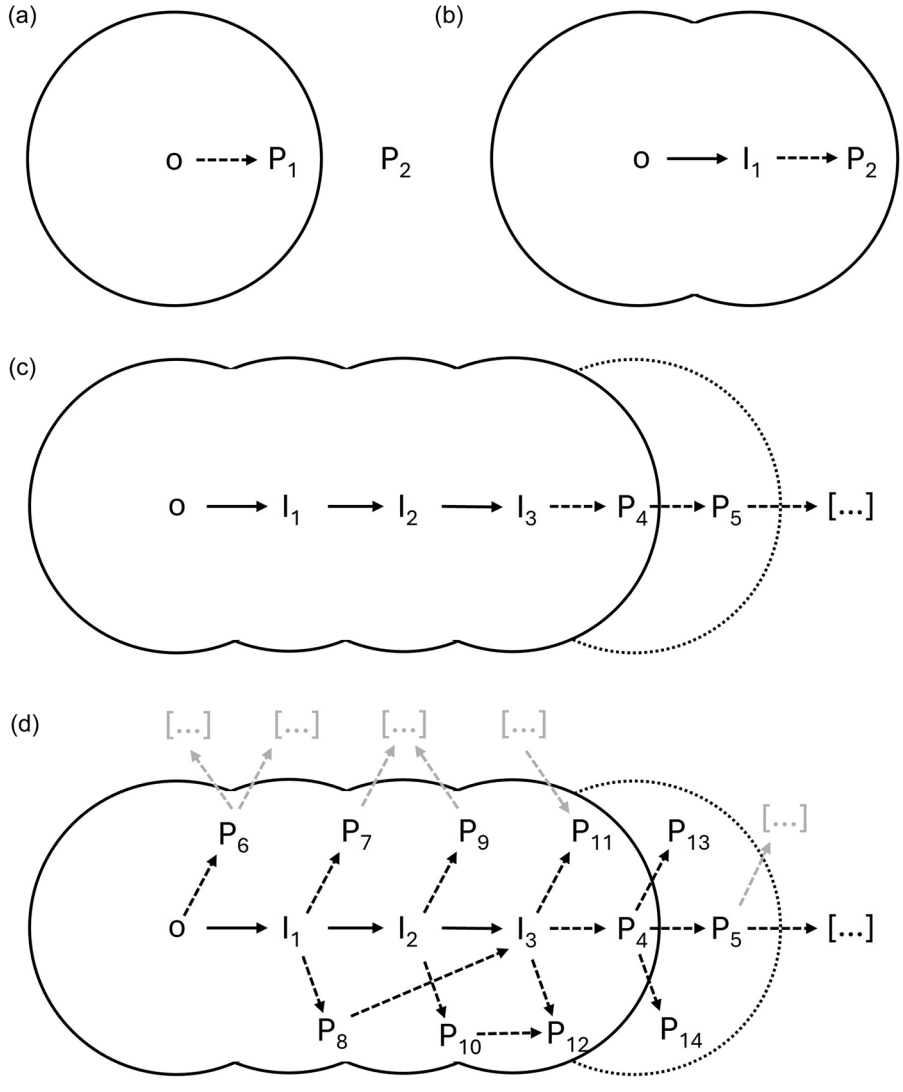


Fig. 1. Expanding reach of invention problem-solving. Full circle denotes a range of problems that can be solved by inventor. o represents core cognitive capacities. P_x represents problems not yet solved. I_x represents problems solved, that is, cognitive technologies providing new cognition operations to inventors. Full circles represent the range of problems an inventor can effectively solve as some time *t*. Dotted circles represent the prospect range of problems an inventor could solve should a new cognitive technology be invented. Full arrows indicate path taken by sequence of cognitive technologies. Dotted arrows point to design problems not yet solved. (A) An inventor, given its available means (operations) has some problems within reach (P_1) and others beyond reach (P_2). (B) Once a new cognitive technology has been invented (I_1), it provides new operations to the inventor, which can now reel in problems that were previously beyond reach (P_2). (C) As new cognitive technologies enrich inventors' repertoire of cognition operations, these reel in within reach the invention of further empowering cognitive technologies, and so on. (D) The process is not linear but branching and probably reticulated, such that some population can take different paths but also converge back on similar problems. In this example, the paths untaken are in gray.

to account for how problem spaces are expanded to include novel, previously inaccessible (unsolvable) problems (Winters & Charbonneau, 2025; Charbonneau et al., 2023). One such way is through the invention of cognitive technologies providing inventors with new, more powerful cognitive capabilities, that is, providing them new cognitive operations for problem-solving.

5. The cognitive empowerment of invention

5.1. Cognitive technologies evolve

In the last 30 years or so, research on the ways by which humans can extend, enhance, or distribute their cognition beyond their brains has formed an extensive literature, one generally known as 4E cognition (Newen, de Bruin, & Gallagher, 2018). Within that literature, some examined how cognitive technologies vary between cultures and across time (e.g., Hutchins, 1995; Sutton, 2010; see also the *History of Distributed Cognition* series, such as Anderson et al., 2019), with fewer examining these in the context of an evolutionary approach to cultural and technological change (e.g., Donald, 1991; Heyes, 2018; Menary & Gillett, 2017). Just what cognitive technologies are, what it means for a technology to be cognitive, and what sorts of cognitive differences these technologies make remain hotly debated. Here is not the place, however, to provide a defense of the metaphysics of cognitive technologies nor of the nature of cognition. For our present purpose, it is enough to adopt some working conceptions of cognitive technologies and the kind of cognitive advantages they provide within the context of invention.

As with most discussions of cumulative technological evolution, I take "technology" as an umbrella term to refer to techniques and artifacts that are acquired through social learning, but also the knowledge required to deploy and use both (and for artifacts, additionally, how to produce, maintain, and repair them). For present purposes, a technology is understood as cognitive when it functions to solve some cognitive problem, that is, a problem for which the solution is the production of some representation (e.g., using a balance scale to *know* which of two objects has the greatest mass). While cognitive technologies are often associated with "higher" cognitive processes such as reasoning, memory, and communication, here technologies serving a perceptual function will be understood as cognitive as well. For instance, X-ray radiography serves as a cognitive technology by producing some representational output (e.g., an X-ray image detecting nascent cavities).

As technologies, cognitive technologies can take several forms. *Cognitive techniques* consist of learned operations and routines for generating, transforming, and processing information so as to solve cognitive problems. For instance, finger reckoning is a cognitive technique helpful to solve simple arithmetic problems. *Cognitive artifacts* consist of material objects that are designed, produced, and used to solve cognitive problems. For instance, a balance scale allows the precise measuring of mass differences between objects or substances. Finally, *representational systems* consist of complexes of representations with their specific internal rules, symbols, and permitted procedures (also representations within that system).

These include number systems and their properties (e.g., Western numbering and the decimal system), writing systems, classificatory and descriptive systems (e.g., the periodic table or the Linnaean system), and so on. Often associated with these are specific procedures (cognitive techniques) directing how to effectively interpret and deploy those systems along their own internal rules (e.g., cross multiplication). The deployment of these systems also generally depends on cognitive artifacts as material means to express them publicly (e.g., through physical inscriptions).

Cognitive technologies—be they techniques, artifacts, representational systems—form traditions diffused and preserved through time and space. These are passed on from one generation to the next through social learning. Cognitive techniques and representational systems are learned from others. For traditions of cognitive artifacts to be passed down the generations, techniques for their production and use (including repair and calibration) must be preserved and learned. Such traditions also vary. For instance, timekeeping technological traditions vary spatiotemporally—be they techniques (e.g., tracking seasonal change in local plants and/or animals or evaluating the orientation and length of shadows), artifacts (e.g., sundials or mechanical clocks), or representational systems (e.g., lunar vs. solar calendars). Moreover, cognitive technologies also change over time through a process of cumulative evolution, with inherited improvements cumulating through time, leading traditions of cognitive technologies to gradually increase in effectiveness. For instance, Huygens' improved Galileo's virge-and-foliot escapement for mechanical clocks, making these clock more precise (Mondschein, 2020).

Because cognitive technologies form traditions that change through time and because they add to the cognitive prowess and repertoire of those who adopt and use them, human cognition is a historically moving target. It is, therefore, useful to make a distinction between historical cognition—those cognitive capabilities that are the result of cognitive technologies—and core cognition—those that are not. Core cognition involves cognitive capabilities that would typically develop (biologically) in the absence of cognitive technologies. In contrast, cognitive technologies are invented and added to the cognitive toolkit of individuals, providing *historically contingent* cognitive capabilities able of solving problems that could not have been solved by individuals relying on their core cognition alone (Section 5.2). Among those problems are invention problems and, of main interest here, those leading to the invention of more powerful cognitive technologies (Section 6).

5.2. Cognitive technologies empower inventive capabilities

For inventors to access problems that were previously beyond reach, a cognitive technology must provide them with new problem-solving capabilities. Cognitive technologies provide new means for problem-solving by effectively adding to the repertoire of cognitive operations one can deploy to solve problems. By inventing a new cognitive technology, inventors effectively provide, to themselves and to others, a new range of cognitive operations, some of which can be used to solve problems previously beyond reach. In other words, cognitive technologies can reel in previously inaccessible (beyond range) problems within the range of solvable problems (Fig. 1B).

Cognitive technologies can enhance an inventor's extant cognitive capabilities (be they core or historical) beyond the bounds of the cognitive processes already available to the inventor. Some technologies allow us to detect and measure quantities more precisely, or beyond the range of what was already detectable and measurable (Humphreys, 2004). Consider the case of measuring the relative mass of two different objects. While our core capacities can manage to discriminate which of two objects with a sufficiently different mass is the heaviest (effectively, discriminating apparent weight), our core perceptual capabilities are imprecise when measuring the relative mass of two objects (Weber's law). In contrast, the invention of the balance scale allowed its users to determine not only which of two objects has the greatest mass, but it also opened up the possibility of precisely comparing their relative mass (used together with the cognitive technique of dividing and comparing the masses). For an inventor, using a balance scale would have helped in figuring out the right quantities of ingredient to produce novel materials (e.g., making different kinds of alloys, glasses), something they could not have reliably done without the scale given the imprecision of apparent hand-weighting alone (Rossi & Russo, 2017). Some cognitive technologies also enhance higher cognitive processes such as memory, for instance, by enhancing the number of items one can remember, or by enhancing how long an item is remembered, and so on. The invention of symbolic means to materially represent quantities and reckon, for instance, would have allowed inventors to produce complex calculations beyond what they could do using their (core) working memory alone. By expanding the bounds of existing cognitive processes, cognitive technologies provide novel problem-solving operations which can then allow inventors to solve design problems they could not have solved without those operations.

If the previous examples provided quantitative changes in pre-existing cognitive capabilities (e.g., by making them broader, more precise, or further reaching), some cognitive technologies also augment the reach of invention problem-solving by providing the inventor with genuinely new cognitive capabilities they did not previously possess. For instance, measuring devices have permitted great improvements not just in precision and range of measurement, but also in the kind of quantities that could be measured. Detection and measurement tools such as the barometer (measuring atmospheric pressure), the compass (measuring electromagnetic fields), and the ammeter (measuring the electric current in a circuit) allowed the perception and measure of natural phenomena inaccessible to our core capabilities. These inventions opened the door to the technological exploitation of phenomena (e.g., electricity and magnetism; [Arthur, 2009]), which in turn could be manipulated to invent novel technologies, including cognitive technologies (e.g., electric computers). In terms of higher cognitive capacities, mathematics and its related techniques have provided a range of operations we could not do otherwise, such as solving general algebraic problems, dealing with conic sections, calculating differentials, and treating statistical abstractions, to name only a few.

It is important to note that these examples are not meant to define some general typology of the range of contributions cognitive technologies provide to inventors in any and all contexts. Instead, these are meant to provide some understanding about how cognitive technologies can empower inventors by bringing within reach invention problems that, without cognitive

technological aid, could not be solved. Determining just how (and whether) a cognitive technology adds to the cognitive repertoire of inventors, we need to compare the cognitive operations available to the inventor before and after the invention of the cognitive technology. This means that the nature of the cognitive contribution provided to an inventor by a new cognitive technology depends on which core and historical means were already available to the inventor.

6. The open-ended technological bootstrapping of invention

Key to the evolutionary dynamics proposed here, the invention of some cognitive technologies makes possible the invention of *some other* cognitive technology the design problems of which were previously beyond reach. These new cognitive technologies, in turn, provide new cognitive capabilities (new cognitive operations), which can be used to invent more cognitive technologies that were, until then, beyond reach. By reeling within reach the invention of more empowering cognitive technologies, inventors can further expand their problem-solving abilities, in turn being capable to reach further, more empowering cognitive technologies. And so on and so forth, in an open-ended manner (Fig. 1C). For short, open-ended technological evolution is the result of a positive feedback loop by which our species is making inventors smarter at inventing cognitive technologies that make them even smarter.

The bootstrapping process described here aims at a different explanatory target than does cumulative optimization. As discussed above (see Section 2), cumulative optimization is a process that allows human populations to adapt to their local environments; it explains *adaptation* through a hill-climbing process of cumulative improvements. In contrast, the bootstrapping process developed here focuses on the expansion of the diversity and complexity of problems that can be solved, that is, it aims to explain the *evolvability and open-endedness* of human technology (O'Brien & Lala, 2023). It does so in two ways: (1) as an explanation of the evolvability of human technology—that is, the capacity for providing new (cognitive) operations and accessing new and more complex problems through cognitive technologies—and (2) as an explanation of the (cultural-technological) evolution of this capacity for evolvability—that is, the continuous expansion of inventive capabilities and the range and complexity of design problems that can be solved, including that of inventing new, more powerful cognitive technologies. As discussed in the previous section, new cognitive technologies can bring within reach problems that were previously out of reach, allowing the population to now search for solutions to these new problems (e.g., through a process of cumulative optimization). In this sense, the invention of new cognitive technologies makes human technological evolution more evolvable as it allows a larger diversity of problems to be solved and, therefore, to produce a greater variety of adaptive solutions. Moreover, because such cognitive technologies also affect the invention process by expanding what can be further invented, the bootstrapping process also provides an explanation as to how the evolvability of human technology can itself evolve: by making inventors increasingly smarter at invention.

To further clarify the workings of this bootstrapping process and its differences to cumulative optimization, we can schematize it as we did cumulative optimization in Section 2:

- OE1 An individual uses its core capabilities to invent a new kind of technology that enhances and augments their problem-solving capabilities, providing a new means to solve new problems, including invention problems. (invention from scratch)
- OE2 Knowledge about how to produce and use the technology is then passed on to others, and further down the generations, forming a tradition. (descent)
- OE3 Downstream, another individual picks up the now traditional technology and uses the novel operations it provides to invent a new technology that they could not have invented had they only relied on their core capabilities. The invention allows for the solving of new problems, including invention problems. (invention through historical cognition)
- OE4 The resulting novel technology is then passed on to others and down the generations. (descent with invention)
- OE5 The process of invention and descent is repeated (OE3 and OE4), but with the operations provided by the new technology (designed in the reiterated step OE3) then used to invent new kinds of technologies that the inventor could not have designed otherwise using their core and previously available historical means, each time allowing to solve new problems that previously could not have been solved, including invention problems. (open-ended evolution)

While providing a detailed case study of this evolutionary process goes beyond the aim of this paper, a simplified sequence is enough to point at the plausibility of the model proposed here. Consider that the empirical observations of geometrical regularities produced by ancient Egyptians using ropes were first made possible by inventing cognitive techniques using those artifacts as surveying instruments (e.g., as straightedges and compasses [Bunt, Jones, & Bedient, 1988]) rather than as mechanical tools. Such cognitive techniques could then be idealized into specific abstract operations (e.g., Euclidean constructions) and, joined with deductive reasoning norms (Dutilh Novaes, 2020; Netz, 1999), made the invention of the Euclidean system possible. In turn, the Euclidean system opened the way for the invention of countless more mathematical and scientific tools we could not have devised otherwise. Among those inventions were previously inaccessible cognitive technologies, such as measuring and calculating instruments (e.g., the theodolite, used in surveying but also in meteorology), sophisticated cognitive techniques (e.g., triangulation), and representational systems (e.g., conic sections and analytical geometry), each capable of solving cognitive problems well beyond our core, unaided cognitive capabilities. And again, these cognitive technologies further expanded the reach of the invention process toward ever more empowering cognitive technologies.

The capability to invent novel, previously unreachable cognitive technologies need not be a direct result of the introduction of a new cognitive technology, however. The bootstrapping dynamic provided here can be indirect, in that a novel cognitive technology can make possible the invention of a noncognitive technology, with only the invention of the latter providing inventors with the means to design a novel, more empowering cognitive technology.

For instance, the invention of the scale allowed the invention of alloys and glasses, which in turn were used to invent glass mirrors. Once glass mirrors were invented, these open the door to inventing novel cognitive technologies requiring such mirrors as parts (e.g., using mirrors in optical instruments such as microscopes).

Such series of inventions, building upon one another, are not increments in the cumulative optimization sense as they are not improvements over existing solutions to a same problem. Going back to our previous example, the cognitive use of ropes for geometrical purposes, leading to geometrical constructions and further to the Euclidean system, is not a story of increasingly lighter, stronger, more durable, or more stretchable ropes. Nor could the problems solved by the later cognitive technologies be solved using the previously available cognitive technologies upon which their invention relied. Indeed, ropes can be used to draw ellipses, one kind of conic section, but hyperbolae and parabolae, having infinite lengths, would require an infinitely stretchable rope to be drawn, a physical impossibility. Instead, the abstract operations forming the mathematics of conics had to be invented to deal with such infinite curves. Such series are about the expanding range of problems one can solve, including the invention of further, more empowering cognitive technologies, not about improving existing means to better solve existing problems.

This is also true of specific steps in the historical, evolutionary process of the series. Zooming in, using ropes to attach to and move objects are noncognitive techniques solving mechanical problems, whereas (later) using ropes as metrological tools (e.g., to re-determine the boundaries of fields along the banks of the Nile after a flood) serve as cognitive techniques enabling the solving of surveying and coordination problems (Bunt et al., 1988). Such inventions—here that of a cognitive technique leading to a change in function (or exaptation) of what was first a noncognitive artifact—are not within the purview of cumulative optimization, as cumulative optimization explains how, given a problem to be solve, populations gradually improve upon solutions to such problem. Something more is required (see Section 2), and that is the capacity to access new problems, a mechanic not considered in the cumulative optimization process (Charbonneau et al., 2023; Winters & Charbonneau, 2025).

The example used here is just that, an example. Several key steps have been simplified to offer a manageable story that illustrates the evolutionary mechanism proposed here. Indeed, there are a lot of advances in geometry and physical tools (e.g., rulers) that could have been included within this story. Moreover, and again for the sake of simplification, the story and the overall evolutionary narrative offered so far as been presented as somewhat linear. However, it is important to keep in mind that there is not one path that was taken throughout history in the range of problems addressed by inventors, nor in the range of cognitive tools that one can develop and solve these. Instead, some populations can come to reach different problems through the same means, or access the same problems using different means, and so on. The road to open-endedness is likely a branching, path-dependent one, with early choices leading to different range of problems to be solved, and all possible paths are likely intermeshed in a reticulated network of accessible but not accessed problems (Fig. 1D). Just how such a network can be mapped and studied, however, is an empirical question best addressed elsewhere.

7. Conclusion

In this paper, I provided a new mechanism of culture-culture coevolution, one examining how our inventive capabilities are empowered by the invention of cognitive technologies. One novelty is to understand invention as a problem-solving process that can be enhanced by the invention of new operations for problem-solving. I argued that cognitive technologies can provide such new operations and in turn allow us to access problems that would be beyond reach otherwise. Among those problems are those of the invention or design of further cognitive technologies that empower inventors with ever more cognitive operations. As this process reiterates through intergenerational time, this allowed our species to display an open-ended process of technological evolution by which we enable ourselves to access and solve increasing complex problems.

The framework developed here demands further expansion. One such line consists of examining what factors may promote or hinder populations to continuously produce empowering cognitive technologies in different populations (Winters & Charbonneau, 2025). Moreover, historically, we also observe societies to develop in an open-ended manner for some time (the so-named "golden ages") but then eventually stall if not decline in the sophistication of the problems they address and the technologies they deploy to do so. There seems to be a great deal of historicity here, which calls for a deeper and finer-grained study of the open-ended process than the one provided here. Further, I have suggested that which cognitive technologies are invented and which problems are and are not accessed may induce path-dependency and even possibly some form of generative entrenchment (Schank & Wimsatt, 1986)—how cognitive technologies may "lock-in" and imper the further development of inventive capabilities (Arthur, 1994)—, thereby acting as constraints on open-endedness rather than as enablers only. All and all, the evolutionary process examined here opens a promising new way to study technological change as an open-ended process, focusing not merely exclusively on solutions and their transmission (see Sections 2 and 3), but by taking seriously how we access and solve ever more complex problems by culturally and technologically empowering our inventive capabilities.

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